Nuclear Instruments and Methods in Physics Research A 607 (2009) 271–278

Contents lists available at ScienceDirect

Nuclear Instruments and Methods in Physics Research A

journal homepage: www.elsevier.com/locate/nima

Distillation of *HD* gas and measurement of spin–lattice relaxation times ☆

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ARTICLE INFO

Article history:
Received 4 July 2008
Received in revised form
9 April 2009
Accepted 18 April 2009
Available online 27 May 2009

Keywords:
HD
Deuterium Hydrid
Nuclear-polarized target
Distillation
Spin–lattice
Relaxation times

ABSTRACT

The polarization (static or dynamic) of *HD* material requires very pure *HD* samples, with H_2 and D_2 impurities concentrations smaller than 0.1%. A new distillation apparatus equipped with a mass spectrometer has been built. It allows reaching such a level of purity and also measurement of the H_2 and D_2 residual concentrations down to 0.05%. The NMR spin–lattice relaxation times T_1^H and T_1^D have also been measured. The apparatus is described as well as the distillation method. T_1^H and T_1^D data as a function of $[H_2]$ and $[D_2]$ concentrations are given, at 0.85 T and 1.8 K.

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1. Introduction

Polarized *HD* targets are made of nearly pure solid molecular *HD* and have an outstanding "Dilution Factor", since all nuclear species in the target are polarizable [1]. The polarization of *HD* molecules has been attempted by both static and dynamic methods, since quite a long time. In 1967, Honig [2] proposed a "Brute Force" (BF) method by which the equilibrium polarization of *H* and *D* nuclei produced at very low temperature and high field (10 mK and 15 T) could be preserved at moderate temperatures and low fields for experimental use. Long relaxation times at experimental conditions (0.2–1.5 K and 1 T) are obtained by letting the ortho-H_2 (o-H_2) and para-D_2 (p-H_2) impurities necessary for the initial static polarization process decay to their magnetically inactive forms para-H_2 (p-H_2) and ortho-D_2 (o-D_2). This ageing at low temperature and high field takes time, but months of ageing can increase relaxation times sufficiently [1] that recently, successful nuclear physics experiments with polarized *HD* targets could be performed [3]. In 1973, Solem [4] investigated the Dynamic Nuclear Polarization (DNP) in radiation-damaged solid *HD*, reaching 3.7% proton polarization and 0.3–0.4% for the vector polarization of deuterons. More recent attempts could not improve the results obtained by Solem [5].

In both cases, it is important to control the purity of the *HD* samples. For the BF method, the polarization of "*H*" requires sizeable H_2 concentrations of the order of 5×10^{-4} to reach equilibrium proton polarizations close to 90% at very low temperature and very high field. The static polarization of "*D*" is limited by the small magnetic moment of the deuteron anyway, so that a vector deuteron polarization higher than 30% is achieved through the transfer of the proton polarization to the deuteron by "Forbidden Adiabatic Fast Passage", a process similar to DNP [3]. Therefore, the ideal *HD* sample for BF polarization should have some H_2 with a concentration of the order of 5×10^{-4} and just enough D_2 to optimize the deuteron polarization process described above, typically $1-5\times10^{-4}$. In particular, in the ageing phase during which o-H_2 decays to p-H_2 and p-D_2 to o-D_2, the decay time of D_2 in *HD* is three times longer than that of H_2, requiring prohibitive ageing delays to get rid of the p-D_2, if too much D_2 had been left in the *HD* sample. In the DNP process, the polarization of free electrons created in the sample is transferred to the nuclear species by RF. This takes place at low field (2.5 T) and "high" temperature (1 K), conditions at which the equilibrium electron polarization is already higher than 90%. To produce high polarization of the nuclear species, their relaxation times T_1^H and T_1^D must be long compared to the electron one T_1^e [6]. At the DNP temperature and field conditions, T_1^H and T_1^D of the order of several minutes can only be reached with H_2 and D_2 concentrations smaller than 2×10^{-4} otherwise, some ageing must be performed before attempting the DNP. Therefore an *HD* sample suitable for DNP should contain as little as possible of H_2 and D_2, at the limit of the distillation performance.

☆ Research supported in part by the European Community through the I3HP Network for the JRA8 Activity.

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doi:10.1016/j.nima.2009.04.055

The distillation of *HD* samples has been performed successfully since quite a long time. Back in the 1960–1970s, relaxation time measurements which were carried in very pure *HD* [7–9], mostly at temperatures between 0.4 and 4.2 K, and fields below 1 T, show that a purity of the distilled *HD* corresponding to $[H_2]$ and $[D_2]$ concentrations in the range of 5×10^{-4} could be achieved, with glass distillation stills in which the cold source was provided by a large reservoir of liquid H_2. This limited the duration of a single distillation and by consequence the amount of *HD* which could be distilled.

At Syracuse University, in the 1980s, the first practical application of very pure *HD* for the production of polarized targets was initiated in order to study nuclear fusion by inertial confinement [10]. More recently, a protocol to produce sizable polarized *HD* targets for nuclear physics experiments has been worked out [11]. This required several moles of purified *HD* and the construction of a new distillation apparatus, using a cryogenerator to liquify the H_2 of the cold source [12]. The production of large *HD* samples having final $[H_2]$ and $[D_2]$ as low as 3×10^{-4} became possible. It could then be demonstrated for the first time, using the ageing technique described above, that very long relaxation times at low field and high temperature could be achieved for sizeable targets. All these studies built up a phenomenological map of the T_1^H and T_1^D dependences on $[H_2]$ and $[D_2]$ in a range of temperatures and fields. However, large uncertainties persisted concerning the $[H_2]$ and $[D_2]$ of a given *HD* sample, because concentrations resulting from a good distillation were usually below the sensitivity of mass spectrometers.

In this paper, a new distillation apparatus using a stainless steel column directly coupled to a cryogenerator cold head is described. In addition, a sensitive mass spectrometer integrated to the system can measure online low $[H_2]$ and $[D_2]$ concentrations, giving the possibility of monitoring the distillation process. T_1^H and T_1^D could then be measured by NMR techniques, allowing for the first time a reliable determination of the spin–lattice relaxation time dependences as a function of H_2 and D_2 impurities in *HD*, since all quantities can be measured independently.

2. Apparatus

Differences in vapor pressure[1] between H_2, *HD* and D_2 can be used to purify *HD* through a distillation process in a rectification column [12]. Successive liquefaction and vaporization along the column will increase the concentration of the lightest element at the top of the column allowing, in a first step, the extraction of H_2 from *HD* and, in a second step, the extraction of pure *HD* from the remaining $HD-D_2$ mixture.

Our apparatus consists of a stainless steel column, based on the geometry of previous glass designs [13] (measuring 30 cm in length and 5 cm in diameter, see Fig. 1), filled with a 20-stage Stedman packing. The cells are separated by a 1 cm thick epoxy ring. The Stedman cells are made of two semi-spherical stainless steel grids fixed on their side like a lens (see Fig. 1). The bottom of the column (the boiler) is a copper pot which contains as much as 10 mol of liquid *HD* and the top (the condenser) is connected to a cryogenerator cold head.

The apparatus includes platinum and carbon temperature probes and heaters fixed on both the condenser and the boiler. The two heaters can each provide up to 6 W of input power. In addition, three capillaries are connected to the bottom, middle and top part of the column in order to extract gas samples at different stages of the rectification process.

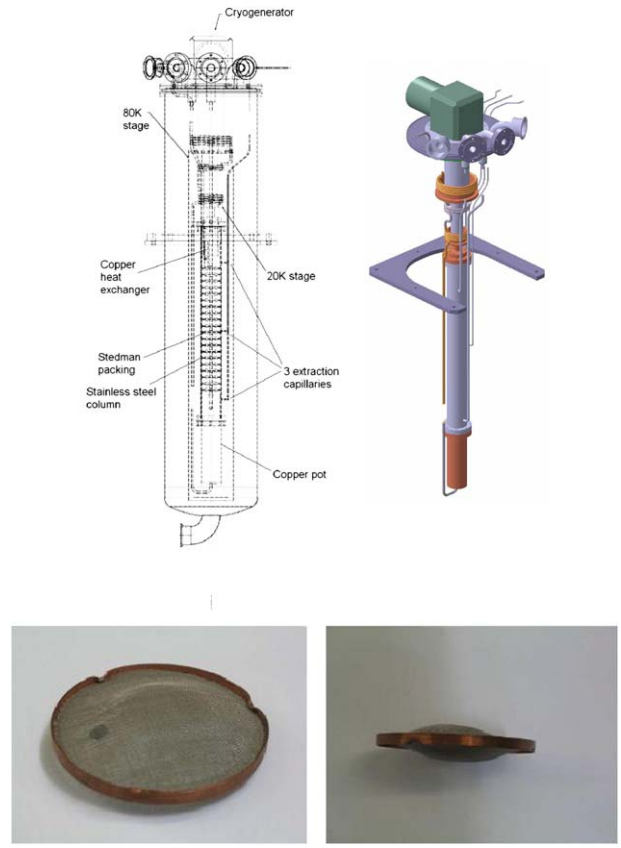

Fig. 1. Design and 3D view of the rectification column. From top to bottom: the condenser cryogenerator, the stainless steel column and the copper pot. We also see the three extraction capillaries. The column is filled with 20 Stedman cells shown at the bottom of the figure.

Concentrations are measured by a commercial "MKS-Spectra Microvision Plus" quadrupole mass spectrometer having a limited mass range (only 1–6 mass units). We have added an input gas manifold designed to guarantee no mass segregation during gas transport from the working pressure of the distillator (500 mbar) to the working pressure of the spectrometer (1×10^{-6} mbar). D_2 concentration can be measured down to 10^{-5} and H_2 down[2] to 2×10^{-4}.

A set of 10 empty tanks (22.8 l each) are connected to the output of the distillator (from one of the three capillaries) via a mass flow controller in order to extract the H_2 or *HD* rich samples. The extraction flow ranges from 1 to 20 ml/min. Finally, an empty tank of 140 l is connected to the column with a check valve (cracking pressure of 2.5 bar) for obvious security reasons. The overall system is controlled by computer via a LabView interface.

3. Distillation procedure

We have distilled 6.2 mol of *HD* gas containing 0.5% H_2 and 0.65% D_2 which is typical commercial "pure" *HD*. The control of the distillation is achieved by the two heaters providing the input power P_c (P_b) on the condenser (boiler). We control the reflux of liquid and vapor in the column by tuning P_b and the working

[1] At 18 K, the vapor pressures for H_2, *HD* and D_2 are, respectively, 461.2, 235.2 and 116.3 mbar, leading to a vapor pressure ratio of order 2 between *HD* and its impurities H_2 and D_2. A more convenient concept used in distillation design is "Relative Volatility", which in our case should be close to the ratios of pure constituents vapor pressure (see Ref. [14] for more details).

[2] Limitation is due to the dissociation of *HD* molecules leading to a significant background at mass 2 from D^+ ions.

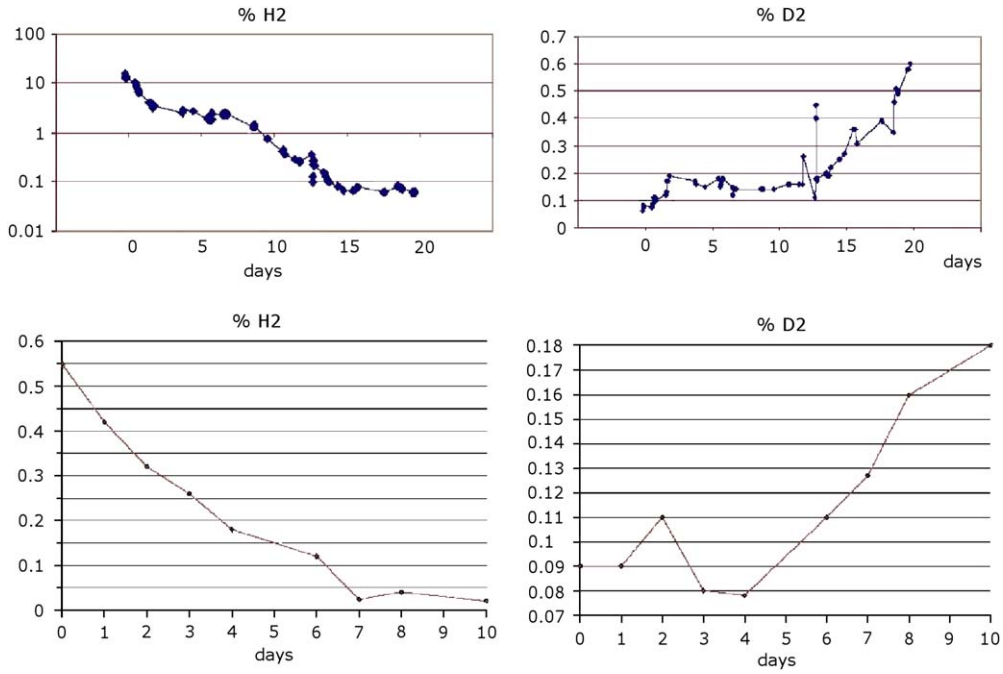

Fig. 2. Evolution of H_2 and D_2 concentrations of extracted gas samples. Top: for the first distillation with an initial gas containing 0.5% H_2 and 0.65% D_2 (different extraction flows were tested, ranging from 1 to 10 ml/min, days 2–6 correspond to 1 ml/min, high fluctuations on day 13 are due to changing running conditions in the distillator), note the logarithmic scale for the graph on the left. Bottom: for the second distillation of sample II gas (extraction flow equal to 1 ml/min).

temperature (or pressure) by controlling P_c. We found that the best efficiency is given by the lowest boiling rate ($P_b = 0$ W) and the lowest temperature. This can be understood by the fact that the equilibrium between liquid and vapor is reached very slowly in our distillator (time constant of the order of 6 h) and that vapor pressure ratios increase when the temperature decreases (up to 2.76 at 15 K between H_2 and HD). The concentration of H_2 at the top of the distillator was found to be 15.5% at 18 K when the distillator reached equilibrium and before any H_2 was removed. The Number of Theoretical Stages (NTS), as calculated with the Fenske [14] relation, for a minimum reflux, is $NTS = 4.02$.

Condenser temperature was fixed to 18.25 K (291 mbar) with an H_2 rich sample on top of the distillator having the following concentrations: $[H_2] = 15.41\%$ and $[D_2] = 0.08\%$. This gas was extracted to one of the output tanks with a flow maintained below 3 ml/min. H_2 and D_2 concentrations are monitored during the process (see Fig. 2). We ended up with three samples: 1.44 mol of H_2 rich gas ($[H_2] = 2.46\%, [D_2] = 0.157\%$), 3.5 mol of H_2 poor gas ($[H_2] = 0.08\%$, $[D_2] = 0.49\%$, sample II) and 1.5 mol of D_2 rich gas ($[H_2] = 0.06\%, [D_2] = 2.52\%$).

Sample II is then cryopumped back into the distillator for a second distillation. Concentrations of extracted gas are shown in Fig. 2. We have three samples at the end of the distillation: the first one is suitable for BF polarization (BF sample): 855 mmol with $[H_2] = 0.26\%$ and $[D_2] = 0.09\%$, the second has been specifically produced for dynamic polarization (DNP sample): 350 mmol with $[H_2] < 0.02\%$ reaching the detection limit of our spectrometer, and $[D_2] = 0.17\%$ and the rest of the gas is $[H_2] = 0.12\%$ and $[D_2] = 0.5\%$. It is important to note that this distillation has been carried out with dynamic polarization in mind. The extraction strategy between the first and the second distillation can significantly change the concentrations and the residual quantity of final samples. $[D_2]$ of the order of 0.08% (with $[H_2]$ still below 0.08%) would have been possible if sample II were not chosen in the first place with such a low H_2 concentration. One should then perform different distillations for BF polarization (very low $[D_2]$) and proton or deuteron DNP (very low $[H_2]$ and $[D_2]$).

Table 1
H2 and *D2* concentrations for the seven samples as measured by the quadrupole mass spectrometer.

Sample name	H_2 concentration	D_2 concentration
R1	1.55×10^{-4}	3.7×10^{-3}
R2	2.0×10^{-4}	1.7×10^{-3}
R3	2.2×10^{-4}	2.9×10^{-3}
R4	4.2×10^{-4}	1.9×10^{-3}
R5	5.4×10^{-4}	2.2×10^{-3}
R6	6.5×10^{-4}	9.0×10^{-4}
R7	6.6×10^{-4}	1.5×10^{-3}

A second distillation session was performed with the same protocol as described above. We finally produced seven different samples, kept at around 500 mbar in seven stainless steel reservoirs. Concentrations of each sample are shown in Table 1.

4. NMR measurements

For all seven *HD* samples produced (see Section 3), we have measured the proton and deuteron relaxation times.

The measurements were performed in a standard "Oxford Instruments" variable temperature cryostat: by pumping on a l-He reservoir through a Joule–Thompson (JT) expansion valve, the system provides a gas flow reaching minimum temperature at 1.8 K. An added heater placed after the JT valve allows the regulation of the temperature up to 30 K. Samples are cryopumped just above the JT valve in a cylindrical Kel-F insert equipped with crossed coils with rectangular shape (25 × 20 mm) and 10 turns each. The insert contains a maximum amount of 200 mmol of solid *HD*. A superconducting magnet provides a field of 0.85 T with a good homogeneity of 10^{-4}.

The first coil on the insert is connected to a FM generator providing a signal at 36.2 MHz, the Larmor frequency of the proton

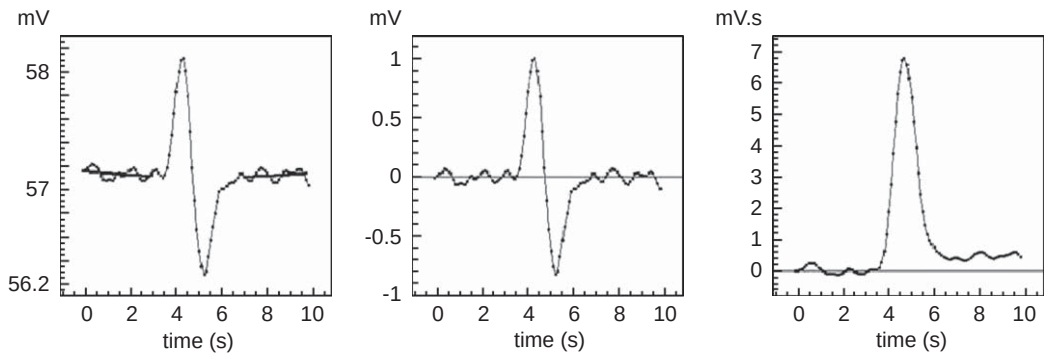

Fig. 3. Proton NMR signal in pure solid *HD*. Left: signal output of the lock-in amplifier. Points are taken every 0.1 s. Field scanning is 4.57×10^{-4} T/s. The sample is probed with a 36.2 MHz and 50 mV excitation signal. Resonance maximum is at 0.85 T. The black line represents the fit of the base line. Middle: NMR signal after baseline subtraction. Right: integrated NMR signal.

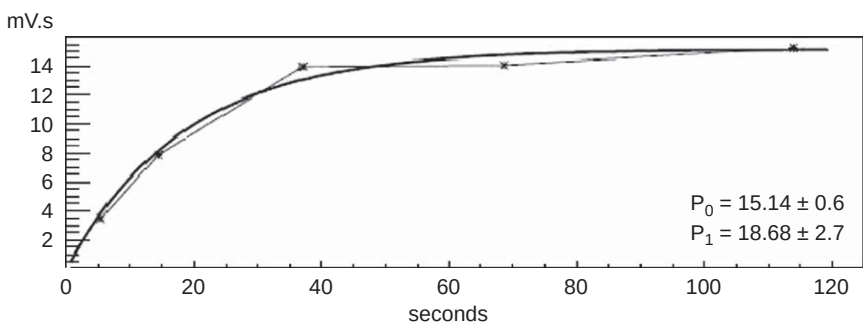

Fig. 4. Polarization recovery curve: proton NMR signal increases after the reset of the polarization. Fit function is exponential: $f(t) = p0 \times (1 - \exp(-t/p1))$.

at 0.85 T, with a frequency modulation of 10 kHz amplitude and 170 Hz frequency (for the deuteron, we set the generator to, respectively, 5.57 MHz, 1 kHz and 170 Hz). The quality factor of this system is Q-factor = 13. The amplitude of the signal is set between 30 and 50 mV. The second coil is connected in series to an impedance matching RLC circuit, an amplification circuit, a demodulation diode, an audio band filter centered at 170 Hz and, finally, to the input of a lock-in amplifier. The lock-in is set to measure a 170 Hz signal and also provides the reference signal at 170 Hz for the FM generator.

The measurement is made in three steps. First, we destroy the equilibrium polarization (at 0.85 T) of the *HD* sample by applying a signal of 1 V at the Larmor frequency of the proton (or deuteron). Second, we measure the polarization by scanning the magnetic field at 4.57×10^{-4} T/s, keeping the FM excitation frequency fixed. Fig. 3 left shows an example of the measured signal. After suppression of the baseline (Fig. 3 middle) and integration (Fig. 3 right), the polarization amplitude is defined as the area under the integrated signal. Finally, we repeat this procedure several times to obtain the polarization recovery curve. Fig. 4 shows this curve along with a simple exponential fit. The relaxation time is then defined as the time constant T_1^H or T_1^D of the fit. When the relaxation time is too long to obtain the full recovery curve, the relaxation time is calculated from the exponential slope (η) at $t = 0$ and the equilibrium amplitude (A_{max}) with the simple relation $T_1 = A_{max}/\eta$. In this case, we get the maximum amplitude by letting the system reach equilibrium overnight or after a few days before each measurement.

5. Results and discussion

Tables 2, 3 and Fig. 5 show the spin–lattice relaxation times T_1^H and T_1^D, at 0.85 T field and 1.8 K temperature conditions, as a

Table 2
Relaxation time T_1^H data table.

Sample name	Ageing time (h)	[o-H_2]	[p-D_2]	T_1^H (s)
R1	17.13	1.04E−04	1.19E−03	46
R2	0	1.50E−04	5.67E−04	383
	62.4	9.99E−05	4.91E−04	1536
	86.4	8.55E−05	4.65E−04	2204
	111.12	7.28E−05	4.39E−04	3018
	134.4	6.25E−05	4.16E−04	4144
	161.8	5.23E−05	3.91E−04	7668
	206.16	3.92E−05	3.53E−04	8873
	228.0	3.40E−05	3.36E−04	13 637
	323.76	1.82E−05	2.70E−04	23 820
	408.0	1.05E−05	2.23E−04	49 320
R3	1.6	1.63E−04	9.63E−04	22.75
	40.8	1.26E−04	8.80E−04	84.97
	62.72	1.10E−04	8.37E−04	173.4
	72.0	1.03E−04	8.20E−04	184.5
	86.26	9.41E−05	7.93E−04	298
R4	1.2	3.18E−04	6.32E−04	17.06
	17.2	2.87E−04	6.09E−04	22.9
R5	2.0	4.03E−04	7.30E−04	7.3
	18.9	3.61E−04	7.02E−04	10.76
	42.1	3.11E−04	6.66E−04	13.4
R6	0	4.87E−04	3.00E−04	22.13
	18.72	4.32E−04	2.87E−04	30.9
	64.32	3.21E−04	2.59E−04	90.36
	91.0	2.70E−04	2.43E−04	219.6
	112.1	2.35E−04	2.32E−04	259.8
	134.9	2.03E−04	2.20E−04	454.8
R7	87.4	2.801E−04	4.092E−04	138
	111.0	2.404E−04	3.878E−04	199.75

function of the o-H_2 concentration [o-H_2]. Concentration error bars shown in Fig. 5 come from the typical resolution of our concentration monitoring system which is around ±5%. These error bars are then calculated taking into account the exponential behavior of [o-H_2] and [p-D_2] during the ageing process (see below). T_1^H and T_1^D error bars have been estimated by performing several measurements in two worst case scenarios (for small and long relaxation times) showing a typical precision of ±20%. No systematic errors have been taken into account.

There are seven samples distilled with the resulting normal H_2 and D_2 concentrations [n-H_2] and [n-D_2] given in Section 3, which have been aged at low temperature to reduce the [o-H_2] and [p-D_2] through the decays of o-H_2 → p-H_2 and p-D_2 → o-D_2, for H_2 and D_2, respectively. Since the conversion rates of molecular species having orbital-angular momentum $J=1$ to configurations with $J=0$ results from interaction of the $J=1$ impurities with the nuclear dipole moments of the H and D nuclei of the HD host, the decay is exponential, with time constants τ_H and τ_D of 6.4 and 18.25 days, respectively [12]. Since T_1^H and T_1^Ddepend on both [o-H_2] and [p-D_2], namely the species having molecular orbital-angular momentum different from zero [2], [o-H_2] and [p-D_2] are quoted in the figures. The T_1^H and T_1^D have been measured regularly during the ageing, with the results plotted in Fig. 5, as a function of [o-H_2]. T_1^D is always longer than T_1^H in the concentration range explored here, with a less drastic dependence on [o-H_2]. Around [o-H_2] $= 4\times10^{-4}$, T_1^D is larger than T_1^H by 2 orders of magnitude, while around [o-H_2] $= 1\times10^{-5}$, both relaxation times become comparable. The dependence on [p-D_2] of both relaxation times is also important: at the same [o-H_2] $=4\times10^{-4}$, T_1^H and T_1^D are getting longer by a factor 6, with [n-D_2] going down from 2.2×10^{-3} (sample R5) to 9×10^{-4} (sample R6). Our T_1^H results can be compared with those of Hardy and Gaines (H&G) [7]. However, since the measurements of H&G were made at 4.2 K and 0.205 T (instead of 1.8 K and 0.85 T), by adding small amounts of normal H_2 (n-H_2) to a distilled sample of HD, one has to correct the corresponding values by a factor 1.6 for the field dependence going as $F^{1/3}$, the [o-H_2] concentrations given by 3/4 × [n-H_2] and the temperature dependence, which varies strongly as a function of [n-H_2] as quoted in Ref. [7]. As can be seen in Fig. 5, there is a qualitative agreement with our data. A detailed comparison should be done at similar [p-D_2] values. In the experiment of H&G, the double distilled HD sample had small but unknown initial [n-H_2] and [n-D_2]. Below [n-H_2] $=1\times10^{-3}$, they found that T_1^H depends nicely on [n-H_2] as [n-H_2]$^{-8/3}$. For [o-H_2] values between 10^{-3} and 10^{-4}, we observe similar dependence. However, for [o-H_2] below 1×10^{-4}, the evolution of T_1^H is quenched, as a result of [p-D_2] decaying much slower with time than [o-H_2].

Table 3
Relaxation time T_1^D data table.

Sample name	Ageing time (h)	[o-H_2]	[p-D_2]	T_1^D (s)
R2	0	1.50E−04	5.67E−04	8857
	62.4	9.99E−05	4.91E−04	22 194
	84	8.68E−05	4.67E−04	25 356
	110.4	7.31E−05	4.40E−04	29 490
	134.6	6.24E−05	4.16E−04	36 502
	162.5	5.21E−05	3.91E−04	36 660
	206.2	3.92E−05	3.53E−04	58 200
	228.5	3.39E−05	3.36E−04	61 704
	398.4	1.12E−05	2.28E−04	111 890
R5	2.05	4.03E−04	7.30E−04	589.26
	18.77	3.62E−04	7.02E−04	1098.25
	42.15	3.11E−04	6.66E−04	1289.5
R6	0	4.87E−04	3.00E−04	1500
	19.92	4.28E−04	2.87E−04	3120
	64.56	3.20E−04	2.59E−04	6324
	90.72	2.70E−04	2.44E−04	7860
	112.08	2.35E−04	2.32E−04	11 244
	134.4	2.03E−04	2.20E−04	11 796
R7	87.52	2.80E−04	4.09E−04	5169
	110.94	2.40E−04	3.88E−04	6949

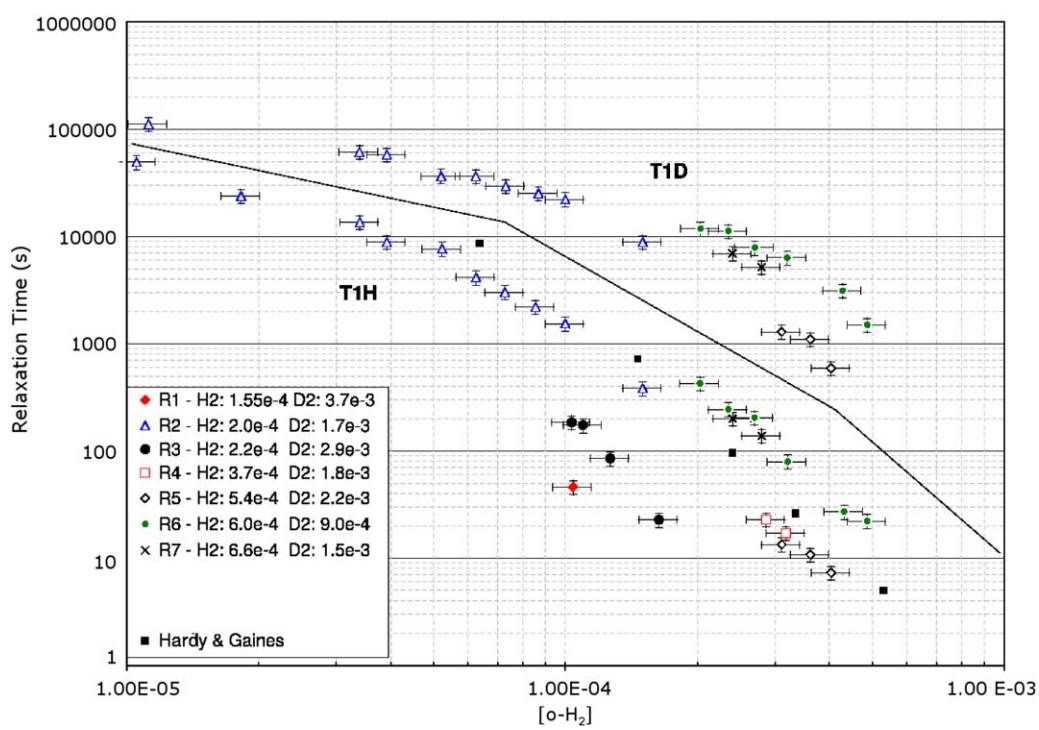

Fig. 5. Relaxation time T_1^H and T_1^D as a function of [o-H_2] for seven different distilled samples. For each sample, initial concentration is given. The black line separates T_1^D data from T_1^H data. T_1^D was measured for samples R2, R5–R7 only and is always higher than T_1^H. Black squares are the data from H&G [7] (see text for details).

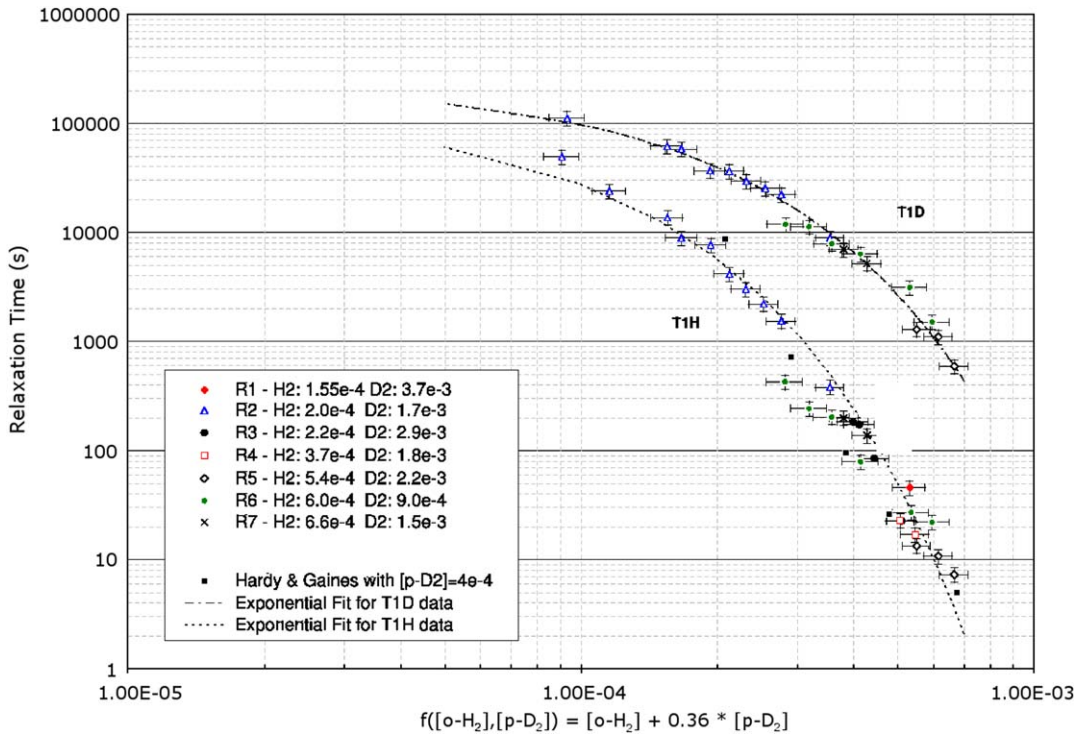

Fig. 6. Relaxation time T_1^H and T_1^D as a function of $f([\text{o-}H_2],[\text{p-}D_2]) = [\text{o-}H_2] + 0.36 \times [\text{p-}D_2]$ for seven different distilled samples (to be compared with Fig. 5). Fits by empirical exponential functions (Eqs. (1) and (2)) are displayed.

Assuming, an initial [n-H_2] of 0.85×10^{-4} for the H&G *HD* sample, as mentioned in Ref. [7], the [n-D_2] of the H&G sample must have been close to 1.2×10^{-3}, which is typical of a good *HD* double distillation aiming at a low [n-H_2]. It is also likely that at very low concentrations of [o-H_2], the relaxation times vary as some inverse power of [p-D_2]. In a previous experiment aiming at the production of polarized targets with long relaxation times, we have found that after 50 days of ageing, which brings [o-H_2] well below 10^{-7} while [p-D_2] remains of the order of 10^{-4} and governs therefore the relaxation process, T_1^H increases exponentially with the number of days, namely with some inverse power of the concentrations [1]. The rising rates increase with lower temperatures, the time constant for T_1^H going from 30 days at 4 K to 15 days at 2 K. This last measurement was made at 2 K and 1 T, in conditions close to those of the present experiment (1.85 K and 1.0 T).

In Fig. 6, we have attempted to express T_1^H and T_1^D as a function of a linear combination of [o-H_2] and [p-D_2]. It turns out that the best agreement, within error bars, is obtained for $[\text{o-}H_2] + 0.36 \times [\text{p-}D_2]$ for both T_1^H and T_1^D. Accordingly one would infer that the depolarization effect of p-D_2 is roughly three times smaller than the o-H_2 one. For T_1^D, the o-H_2 depolarization effect is still dominant as compared to the p-D_2 one. Within the range of concentrations explored here, both curves could be fitted reasonably well by empirical exponential functions of the form

$$T_1^H = (1.34 \pm 0.2)10^5 \times e^{-(1.58\pm0.05)10^4 \times ([\text{o}-H_2]+0.36\times[\text{p}-D_2])} \quad (1)$$

$$T_1^D = (2.38 \pm 0.3)10^5 \times e^{-(9\pm0.4)10^3 \times ([\text{o}-H_2]+0.36\times[\text{p}-D_2])}. \quad (2)$$

Fig. 7 shows our measured T_1^H as a function of [o-H_2] at fixed values of [p-D_2], spanning one order of magnitude from 2.1×10^{-4} till 1.18×10^{-3}. There are few points and a functional dependence is needed to draw conclusions, but at $[\text{o-}H_2] = 10^{-4}$, T_1^H rises from 50 to 1500 s (a factor 30), when [p-D_2] goes down from 1.2×10^{-3} to 5.0×10^{-4}. Fig. 8 summarizes the dependence of T_1^H versus [p-D_2]. When [p-D_2] is 10 times larger than [o-H_2]([o-H_2] below 1×10^{-4}), the dependence can be steep, almost comparable to the dependence versus [o-H_2].

Finally, Fig. 9 and Table 4 show the temperature dependence between 1.8 and 10 K for both T_1^H and T_1^D at typical [o-H_2] and [p-D_2] of 2.5×10^{-4} and 1×10^{-3}, respectively. There is a much steeper dependence on temperatures for T_1^D than for T_1^H. One should keep in mind that, as quoted in Ref. [7], the temperature dependence increases drastically when relaxation times increase.

It has been previously empirically deduced [15] that the functional dependence of T_1^H and T_1^D on [o-H_2] and [p-D_2] at 1.2 K and 0.1 T was the following:

$$T_1^H \propto [\text{o-}H_2]^{-2.1} \times [\text{p-}D_2]^{-1.6} \quad (3)$$

$$T_1^D \propto [\text{o-}H_2]^{-1.0} \times [\text{p-}D_2]^{-3.9}. \quad (4)$$

Our temperature 1.8 K is close enough and the field dependence can be taken into account by a scaling factor. This allows us to make comparisons with our data. However, the functional dependence of T_1^H and T_1^D on [o-H_2] and [p-D_2] in a wide range of concentrations cannot be of the form given by Eqs. (3) and (4). If, for example, [p-D_2] would be maintained at a fixed value as in Ref. [7], T_1^H and T_1^D would tend to their saturation values driven by [p-D_2] when [o-H_2] tends to zero. Accordingly, Eqs. (3) and (4) are only valid in a limited range of concentrations. Figs. 7 and 8 indicate that, for fixed [p-D_2] ([o-H_2]), the dependence of T_1^H on the inverse power of [o-H_2] ([p-D_2]) increases when the fixed concentration goes down and follows the variation of the changing concentration, namely decreases when the changing concentration decreases.

In the range of concentrations explored in this paper, namely 1×10^{-3} to 1×10^{-5}, T_1^H is assumed to be proportional to $[\text{o-}H_2]^{-\alpha} \times [\text{p-}D_2]^{-\beta}$. The concentrations inverse power coefficients α and β can be deduced from the slopes of the curves of Figs. 7 and 8 which give the variation of T_1^H at fixed [p-D_2] and [o-H_2], respectively. One finds that both α and β vary between 1 and 3.5

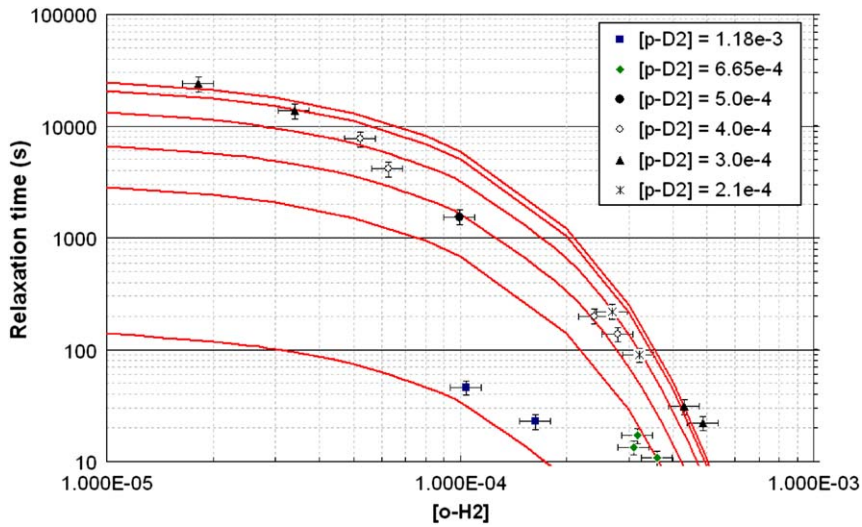

Fig. 7. Relaxation time T_1^H as a function of [o-H_2] at fixed values of [p-D_2]. Lines are calculated from Eq. (1) (see text) for each [p-D_2] concentrations (from 1.18×10^{-3} down to 2.1×10^{-4} from bottom to top).

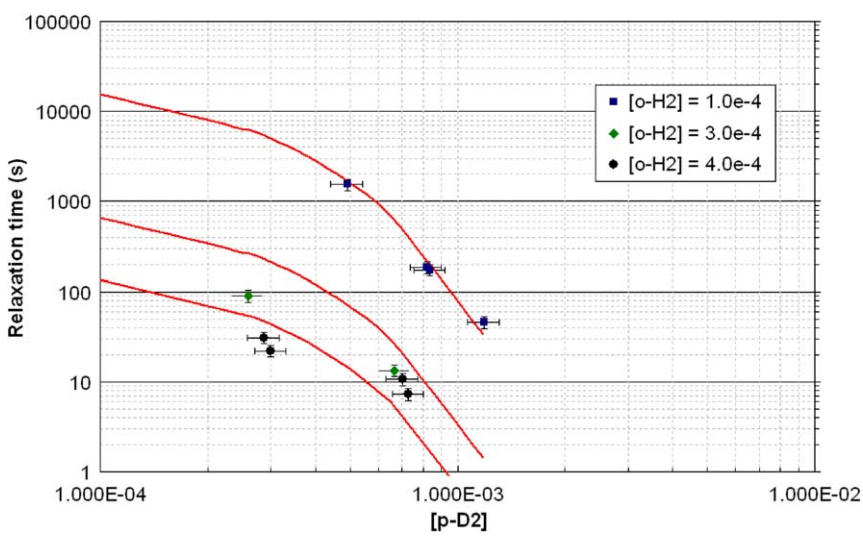

Fig. 8. Relaxation time T_1^H as a function of [p-D_2] at fixed values of [o-H_2]. Lines are calculated from Eq. (1) (see text) for each [o-H_2] concentration (from 4.00×10^{-4} down to 1.0×10^{-4} from bottom to top).

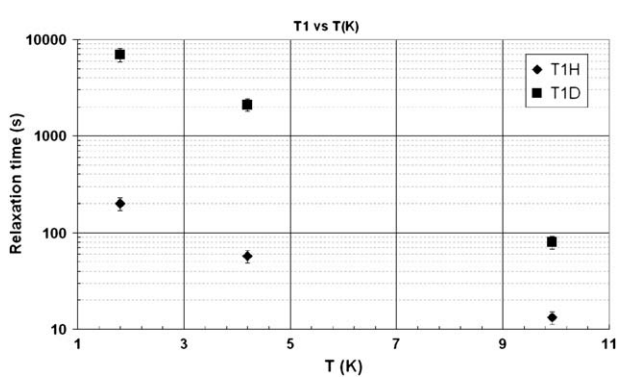

Fig. 9. Relaxation time as a function of temperature for sample R7 (see Table 1).

in qualitative agreement with Eq. (3). An indication of the T_1^H dependence on [p-D_2] can also come from our T_1^H measurements done at 1.9 K and 1.0 T [1]. After 50 days of ageing, T_1^H grows as

Table 4
Relaxation time T_1^H and T_1^D data table.

Sample name	T (K)	Ageing time (h)	[o-H_2]	[p-D_2]	T_1^H (s)	T_1^D (s)
R7	1.8	110.96	2.40E−04	3.88E−04	199.75	6949
	4.2	114.77	2.34E−04	3.84E−04	57.3	2104
	9.93	116.1	2.32E−04	3.83E−04	13.17	79.6

$\exp(0.0632 \times d)$. Since the [p-D_2] decays exponentially with a time constant τ_D of 18.25 days and drives alone the depolarization process ([o-H_2] lower than [p-D_2] by 3 orders of magnitude, so well outside the concentrations range explored here), we get a [p-D_2] power coefficient of $1.15 = 18.25 \times 0.0632$, in reasonable agreement with Eq. (3). Unfortunately, no measurement of T_1^D was done in the experiment of Ref. [1]. Within the precision of our present data, we cannot confirm the steep dependence of T_1^D versus [p-D_2] as given by Eq. (4). Comparing T_1^D for samples R5 and R6, we rather get equal power coefficients close to 1.7 for both [o-H_2] and

[p-D_2] on most of the range. It should be mentioned, however, that in the highest [o-H_2] region (see Fig. 5), the T_1^D dependence on [o-H_2] increases drastically with [o-H_2]. A rule of thumb is that the slopes of the curves giving T_1^H as a function of [o-H_2] (see Fig. 7) become larger when [p-D_2] decreases. The same behavior is observed for T_1^H as a function of [p-D_2] for smaller and smaller [o-H_2] (see Fig. 8). It is anticipated that T_1^D behaves as T_1^H, although our present data do not allow definite statements. To conclude, we can say that the inverse power coefficients of Eqs. (3) and (4) are within a reasonable range, for H_2 and D_2 concentrations explored in this paper.

6. Conclusions

In the present work, a new distillation apparatus has been developed to purify *HD*, allowing residual concentrations of [n-H_2] and [n-D_2] in the range of 10^{-4} to be reached and it has been shown that an online mass spectrometer can measure the concentrations well below 10^{-3}. This performance has permitted for the first time measurement of the dependence of spin–lattice relaxation times T_1^H and T_1^D in solid "*HD*" as a function of both concentrations [o-H_2] and [p-D_2], all parameters being measured independently. The dependence is steep for T_1^H but much less steep for T_1^D. Using ageing and taking into account the decay of [o-H_2] and [p-D_2], concentrations below 10^{-4} could be investigated. In this range of concentrations, T_1^D always remained larger than T_1^H. Finally, we have attempted to quantify the functional dependence of both T_1^H and T_1^D on [o-H_2] and [p-D_2]. We find that o-H_2 plays the leading role in the depolarization process of "*H*" and "*D*". This last behavior is rather surprising and should be confirmed by additional measurements of T_1^D, which are now possible at the Research Center for Nuclear Physics of the Osaka University in Japan, where all the equipment described here has been transferred.

Acknowledgments

During the measurements reported in this work, S. Bouchigny has been hired as postdoc in the framework of the European Community I3HP Network (JRA8 program). It is a pleasure to thank Pr. A. Honig from Syracuse University for many fruitful discussions and plentiful suggestions.

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